

Solutions – SML MidSem Winter 2026

Q1

Estimator:

$$\hat{\mu} = x_1$$

Since samples are i.i.d. with mean μ ,

$$\mathbb{E}[\hat{\mu}] = \mathbb{E}[x_1] = \mu.$$

Hence,

The estimator is unbiased.

Q2

Binary feature vector:

$$x \in \{0, 1\}^d$$

with conditional independence.

Likelihood

Class ω_1 :

$$p(x|\omega_1, \theta) = \prod_{i=1}^d \theta_i^{x_i} (1 - \theta_i)^{1-x_i}$$

Class ω_2 :

$$p(x|\omega_2, \phi) = \prod_{i=1}^d \phi_i^{x_i} (1 - \phi_i)^{1-x_i}$$

Posterior

Posterior is directly proportional to likelihood times prior.

Beta pdf is $\theta^{\alpha-1}(1-\theta)^{\beta-1}/Z$

Let

$$s_{1i} = \sum_{x \in \mathcal{D}_1} x_i, \quad s_{2i} = \sum_{x \in \mathcal{D}_2} x_i$$

Using Beta–Bernoulli conjugacy,

$$p(\theta_i|\mathcal{D}_1) = \text{Beta}(\alpha_i + s_{1i}, \beta_i + N_1 - s_{1i})$$

$$p(\phi_i|\mathcal{D}_2) = \text{Beta}(\alpha_i + s_{2i}, \beta_i + N_2 - s_{2i})$$

Decision Boundary

Discriminant:

$$g(x) = \log p(x|\omega_1) - \log p(x|\omega_2) + \log \frac{P(\omega_1)}{P(\omega_2)}$$

After simplification:

$$w_i = \log \frac{\theta_i(1 - \phi_i)}{\hat{\phi}_i(1 - \theta_i)}$$

$$b = \sum_{i=1}^d \log \frac{1 - \theta_i}{1 - \phi_i} + \log \frac{P(\omega_1)}{P(\omega_2)}$$

$$w^\top x + b = 0$$

Q3 Modified PCA

Mean:

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

Covariance (without explicit centering):

$$\Sigma = \frac{1}{N} (X - \mu)(X - \mu)^\top$$

Projected variance:

$$\text{Var}(u^\top x) = u^\top \Sigma u$$

Constraint:

$$(u^\top \mu)^2 = 1 \Rightarrow u^\top (\mu \mu^\top) u = 1$$

Optimization:

$$\max_u u^\top \Sigma u \quad \text{s.t.} \quad u^\top (\mu \mu^\top) u = 1$$

Lagrangian gives:

$$\Sigma u = \lambda (\mu \mu^\top) u$$

Q4 Modified FDA

Means:

$$\mu_1 = \frac{1}{N} \sum_{x \in X_1} x, \quad \mu_2 = \frac{1}{N} \sum_{x \in X_2} x$$

$$m = \mu_1 - \mu_2$$

Within-class scatter:

$$S_W = S_1 + S_2$$
$$S_k = \sum_{x \in X_k} (x - \mu_k)(x - \mu_k)^\top$$

Total scatter:

$$S_T = \sum_x (x - \mu)(x - \mu)^\top$$

μ is the global mean. $\mu = (1/2N) \sum_x x_i$ Constraint:

$$(w^\top m)^2 = 1$$

Objective

$$\max_w w^\top S_T w \quad \text{s.t.} \quad w^\top S_W w = 1, \quad w^\top m m^\top w = 1$$

Solution

$$S_T w = \lambda S_W w + \gamma m m^\top w$$

or

$$(S_T - \gamma m m^\top) w = \lambda S_W w$$

Q5 PCA

$$X = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$$

Mean:

$$\mu = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Centered matrix:

$$X_c = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$$

Unbiased covariance:

$$\Sigma = X_c X_c^\top = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

Eigenvectors:

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Q6 MAP Estimate

Likelihood:

$$x_i \sim \mathcal{N}(\mu, I)$$

Prior:

$$p(\mu) \propto e^{-\lambda^\top \mu}, \quad \mu_i \geq 0$$

Posterior maximization:

$$\min_{\mu \geq 0} \frac{1}{2} \sum_{i=1}^N \|x_i - \mu\|^2 + \lambda^\top \mu$$

Let

$$\bar{x} = \frac{1}{N} \sum x_i$$

Unconstrained solution:

$$\mu^* = \bar{x} - \frac{\lambda}{N}$$

Applying constraint:

$$\mu_{MAP} = \left[\bar{x} - \frac{\lambda}{N} \right]_+$$

Q7 Trace Ratio

$$\max_W \frac{\text{tr}(W^\top S W)}{\text{tr}(W^\top W)}$$

Check the class notes for derivation.

Q8 FDA Vector

Computation of Within-Class Scatter

Class samples are given as

$$X_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad X_2 = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}.$$

Hence the samples are

$$\begin{aligned} x_{11} &= (2, 0)^\top, \quad x_{12} = (0, 2)^\top, \\ x_{21} &= (2, 0)^\top, \quad x_{22} = (0, -2)^\top. \end{aligned}$$

Class Means

$$\mu_1 = \frac{1}{2} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mu_2 = \frac{1}{2} \begin{bmatrix} 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

$$m = \mu_1 - \mu_2 = (0, 2)^\top$$

Scatter of Class 1 Deviations:

$$x_{11} - \mu_1 = (1, -1)^\top, \quad x_{12} - \mu_1 = (-1, 1)^\top.$$

Outer products:

$$(1, -1)(1, -1)^\top = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix},$$

$$(-1, 1)(-1, 1)^\top = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

Thus,

$$S_1 = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}.$$

Scatter of Class 2 Deviations:

$$x_{21} - \mu_2 = (1, 1)^\top, \quad x_{22} - \mu_2 = (-1, -1)^\top.$$

Outer products:

$$(1, 1)(1, 1)^\top = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix},$$

$$(-1, -1)(-1, -1)^\top = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

Hence,

$$S_2 = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}.$$

Within-Class Scatter

$$S_W = S_1 + S_2 = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} + \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} = \boxed{\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}} = 4I.$$

FDA direction:

$$w \propto S_W^{-1} m = (0, 1)^\top$$

Q9 Bonus

Likelihood:

$$p(D|a, b) = (b - a)^{-n}$$

Maximized when interval is smallest containing data:

$$= \min_i x_i, \quad \hat{b} = \max_i x_i$$